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REV	DESCRIPTION	DATE(YEAR-MO-DA)	APPROVED
1			

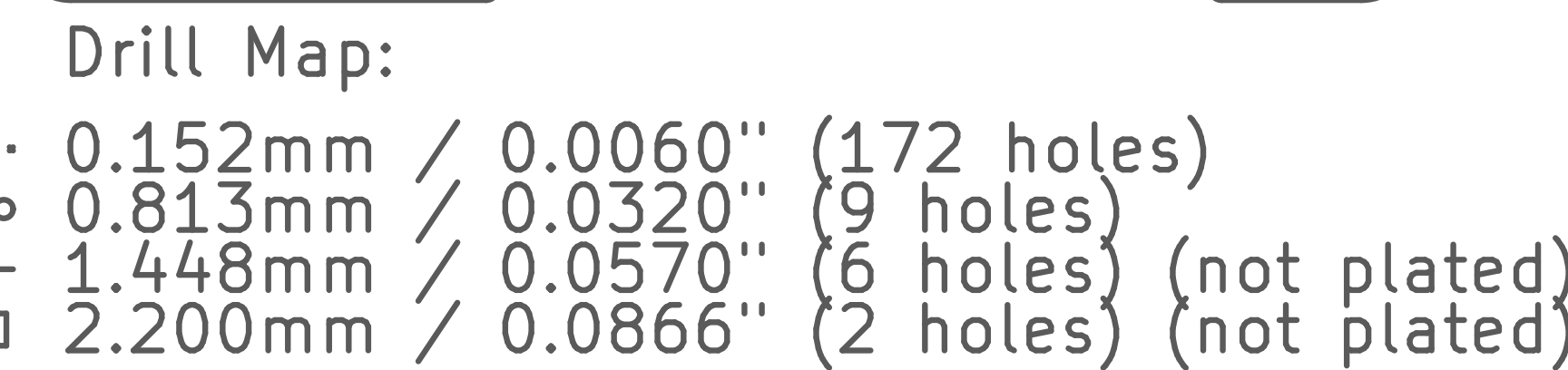
A technical line drawing of a rectangular plate with rounded corners. The plate is oriented diagonally. It features two screws, one at the top-left corner and one at the bottom-right corner. The drawing is a simple black-and-white line art representation.

The schematic diagram illustrates the layout of a printed circuit board (PCB) for a microcontroller-based system. The components are distributed across the board as follows:

- Microcontrollers:** U1 (central), U2 (bottom right), U3 (top right), and U4 (top left).
- Memory:** L1, L2, and L3 (top right area).
- Logic/Decoder:** D1 and D2 (center).
- Peripheral Components:** P1, P2, P3, P4, and P5 (bottom left area).
- Passive Components:** Resistors (R1-R14) and capacitors (C1-C20) are placed throughout the board, often near the microcontrollers and memory modules.
- Other ICs:** IC1 through IC9 are located near the microcontrollers and memory.
- Connectors/Ports:** Q1 and Q2 are located near the bottom left, and Q3 is near the bottom right.
- Board Features:** The PCB has mounting holes at the corners and a large cutout at the bottom center.

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SCALE: 1:1 UNIT WEIGHT: SHEET 1 OF 1



NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING

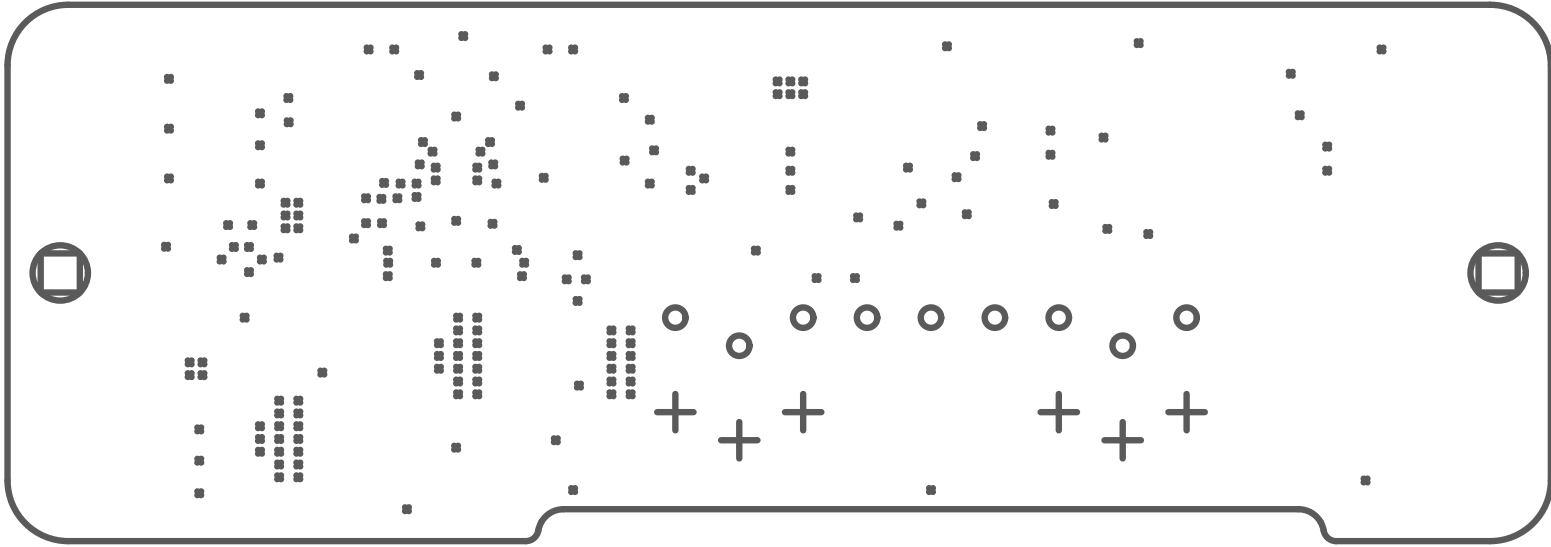
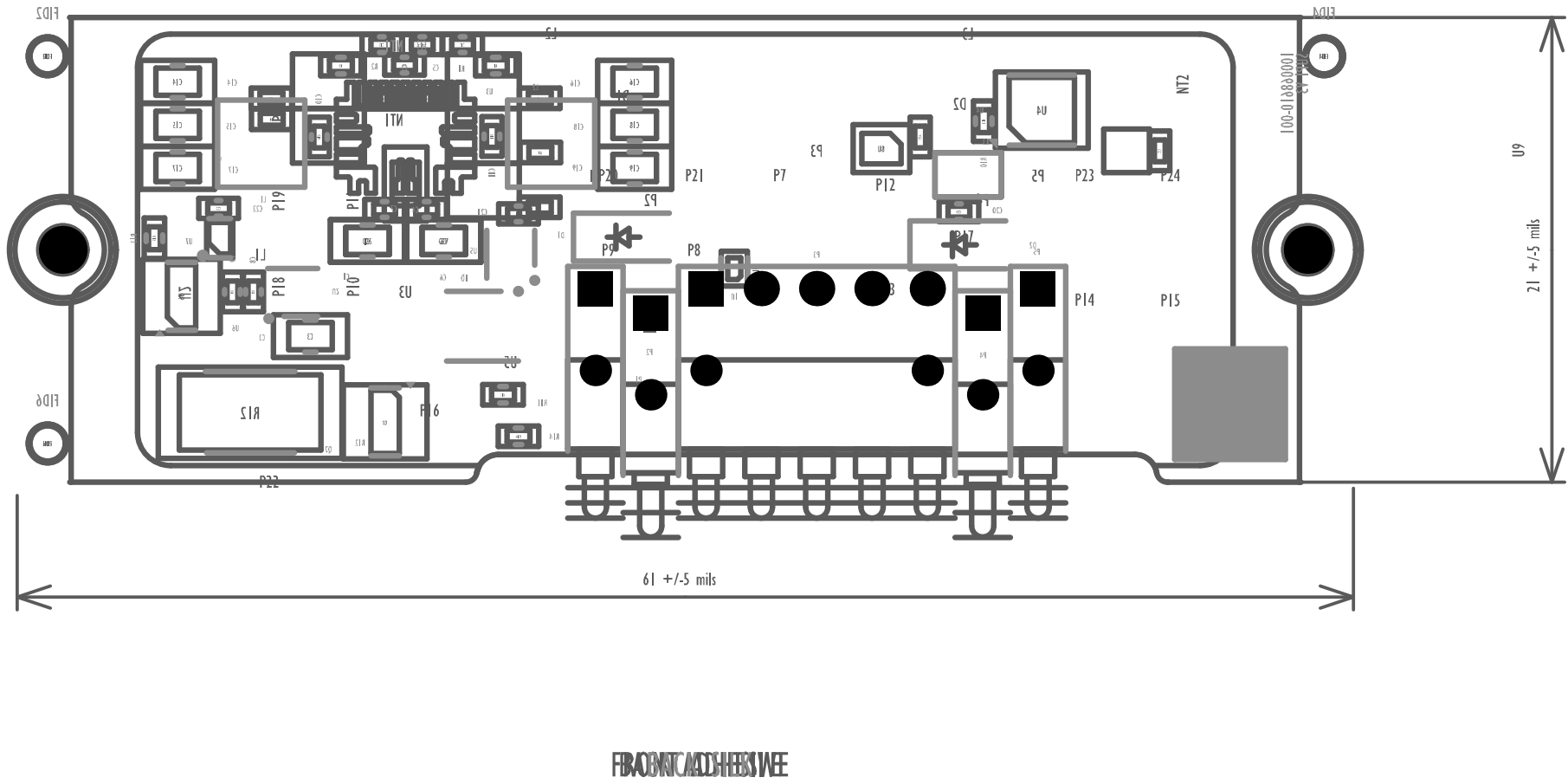
IMPEDANCE					
TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
1	2	90	6.4	4.5	SMT COPLANAR W/ENCLOSURE W/VOID



- | | | | | |
|---|---------|---|---------|------------------------|
| • | 0.152mm | / | 0.0060" | (172 holes) |
| ◦ | 0.813mm | / | 0.0320" | (9 holes) |
| + | 1.448mm | / | 0.0570" | (6 holes) (not plated) |
| □ | 2.200mm | / | 0.0866" | (2 holes) (not plated) |

NOTES: UNLESS OTHERWISE SPECIFIED

1. THE PWB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-6013 CLASS 2.
2. DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
3. TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-650 2.4.2.2.
4. DO NOT EXPOSE TO DRAWING.
5. MATERIAL:
 1. THE PWB CONDUCTIVE MATERIAL SHALL BE 350HR.
 2. THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
 3. FINISHES:
 1. BOW AND TWIST OF ASSEMBLY OUT-PAANEL OR UNGULATED PWB SHALL NOT EXCEED 0.75%.
 2. TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-650 4.4.2.2.
 4. ETCH GEOMETRY:
 1. MEASURE WIDTH FROM THE BACK OF THE METALLIZATION.
 2. INSULATED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERMER PATTERN PITCH BY MORE THAN 20%.
 5. SURFACE FINISH:
 1. FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
6. HOLE SIZE TABLE FOR DIMENSIONS AFTER PLATING (WHERE APPLICABLE):
 1. ALL PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-6012, CLASS 2.
 2. ALL HOLES SHALL MEET MINIMUM HOLE SIZE AND PLATING:
 1. ALL HOLES SHALL BE LOCATED WITHIN 3 MIL OF TRUE POSITION AS SUPPLIED IN CAD DATA.
 2. THE TOLERANCE FOR ALL HOLES SHALL AS INDICATED IN THE HOLE TABLE.
 3. ALL HOLES IN THE HOLE TABLE MARKED AS (copper+clay+epoxy+fill+through) SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
7. SOLDERMASK:
 1. SOLDERMASK OVER BASE COPPER (EMBO) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED NETWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-6540 4.0.1 TYPE B.
 2. COLOR: GREEN.
 3. LIQUID PHOTO-IMAGABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS, HALOGEN FREE.
 4. NO REEF-OUT ALLOWED OVER EXPOSED PADS.
8. SOLDEREN:
 1. SOLDEREN PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-POURTY ENIG INK.
 2. COLOR: WHITE.
 3. ANY UNDESIRED STAGNE WIDTH SHALL BE 5 MIL.
 4. CLIP SOLDEREN AWAY FROM ANY EXPOSED METAL.
9. PREPARE:
 1. REMOVE ALL BURRS AND BREAK SHARP EDGES.
 2. IF NECESSARY, DEBURR WITH #100.
10. USE INSPECTION REFERENCE TABLES:
 1. YENDOR MAY ADD TEST GEOMETRIES UP TO +20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
 2. YENDOR MUST ADJUST SPECIFIC COES OR PREPAGS TO ACHIEVE IMPEDANCES AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
11. PARALLELED PWS LAYOUT:
 1. IF THE PWB IS PARALLELED, THE PWB HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PADS TO ANY LOCATION ON THE PANEL TO EASE FABRICATION ON TEST.
 2. IF THE PWB IS NOT PARALLELED:
 1. THE PWB SHORTS
 2. CONTROL THE NET LINES SHORTS TABLE FOR NETS THAT ARE INTENTIONALLY SHORTED WITH NETTED PADS.
12. NON-STRUCTURAL EVALUATION:
 1. ALL PWBs SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-2463 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-2552, CLASS 2.
 2. IF ANY OF THE COMPONENTS SHALL BE SUPPORTED WITH SOLDER, THE PWB SHALL BE MARKED WITH SOLDER SUPPORT.
13. ALL AREA BORDERS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-HSTD-330, WITH THE LETTERS "T". MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.



Drill Map:

- 0.152mm / 0.0060" (172 holes)
- 0.813mm / 0.0320" (9 holes)
- + 1.448mm / 0.0570" (6 holes) (not plated)
- 2.200mm / 0.0866" (2 holes) (not plated)

NOTES, UNLESS OTHERWISE SPECIFIED

1. THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-4012 CLASS 2.
2. INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
3. TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.3.22.
4. DO NOT SCALE TO DRAWING.
5. MATERIAL
6. THE PCB DIELECTRIC MATERIAL SHALL BE STYRE.
7. THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
8. FLATNESS
9. 1. HOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
10. TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
11. ENIG COATING
12. MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
13. FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
14. SURFACE FINISH
15. FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
16. HOLES
17. 1. SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
18. 2. PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-4012, CLASS 2.
19. 3. SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
20. 4. ALL HOLES SHALL BE LOCATED WITHIN 5 MIL OF THEIR POSITION AS SUPPLIED IN CAD DATA.
21. 5. THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
22. 6. HOLES IN THE HOLE TABLE MARKED AS (000000)CAPPED/FILLED(000000) SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
23. 7. SOLDERMASK
24. 1. SOLDERMASK OVER BARE COPPER (P100C) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ANTIWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
25. 2. COLOR: GREEN
26. 3. LIQUID PHOTO-IMAGABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS; HALOGEN FREE.
27. 4. NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
28. 5. NO BLEED
29. 6. SOLDERMASK PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-AUTIENT EPOXY INK.
30. 7. COLOR: WHITE
31. 8. ANY UNSPECIFIED STRAKE WIDTH SHALL BE 5 MIL
32. 9. CLIP SOLDERMASK AWAY FROM ANY EXPOSED METAL
33. 10. FINISHING
34. 1. REMOVE ALL BURRS AND BREAK SHARP EDGES.
35. 2. IMPEDANCE (ALL TOLERANCED +/- 10%)
36. 3. SEE IMPEDANCE REFERENCE TABLES
37. 4. VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
38. 5. VENDOR MAY SUBSTITUTE SPECIFIC CORES OR PREPREGS TO ACHIEVE IMPEDANCE TARGETS AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
39. 6. PANELIZED PWB LAYOUT
40. 7. IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBs TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
41. 8. NET TIES AND SHORTS
42. 9. CONSULT THE NET TIES SHORTS TABLE FOR NETS THAT ARE INTENTIONALLY SHORTED WITH NET TIES.
43. 10. NON-DESTRUCTIVE EVALUATION
44. 1. ALL PWBs SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-365 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
45. 2. CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
46. 3. ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS "ET". MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.

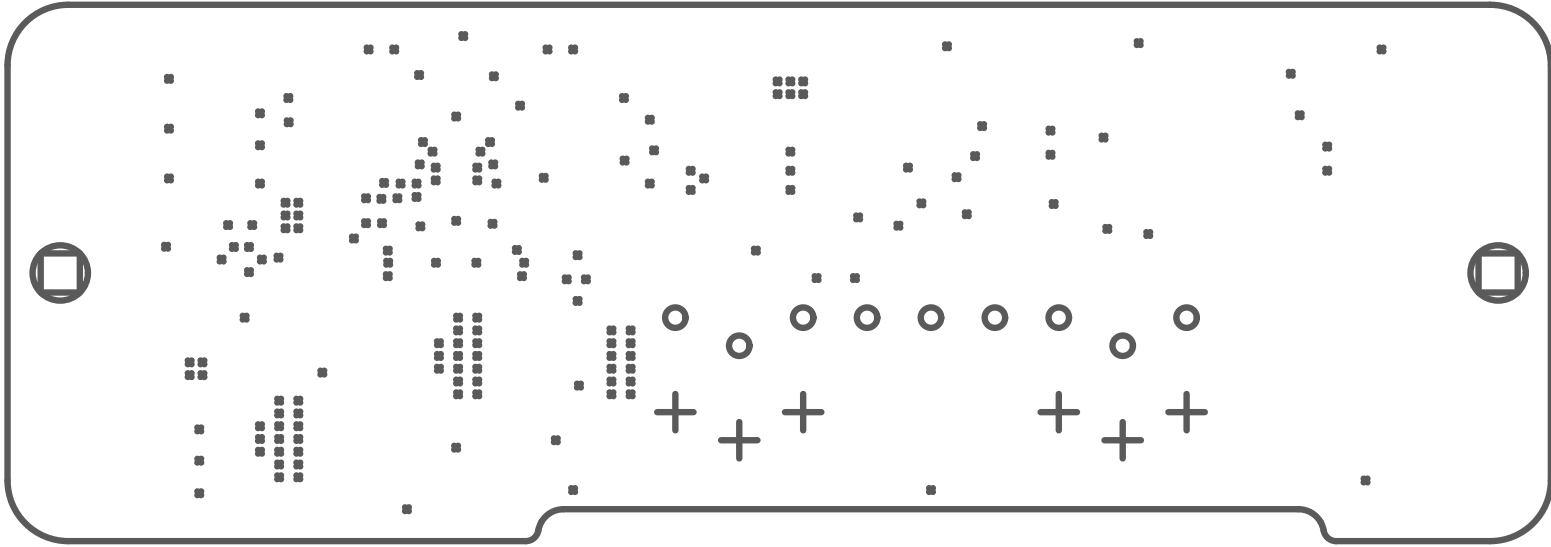
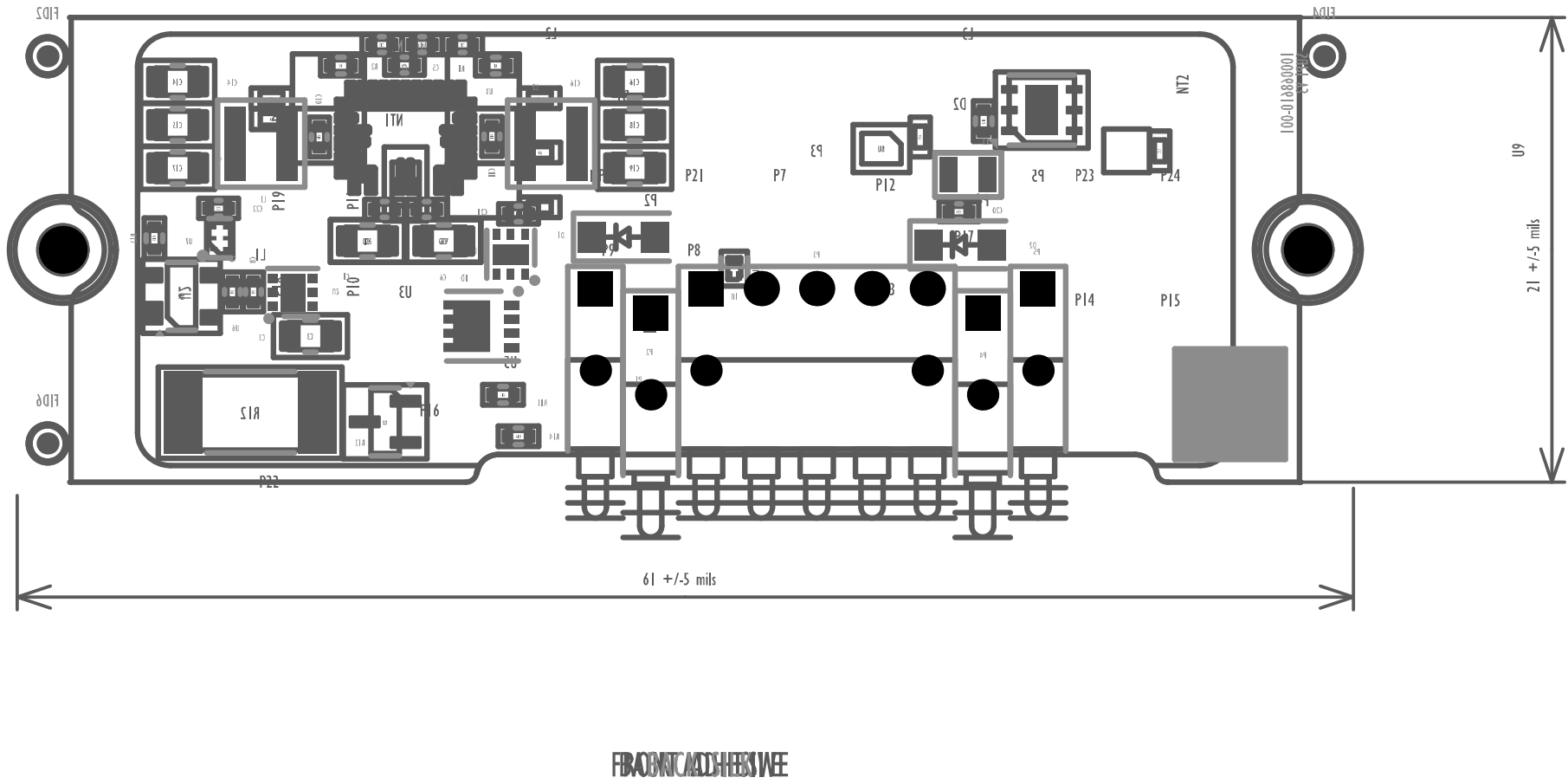
LAYER STACKUP AND FILENAME

LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_1_SOLDERMASK.GBR	String	String
FRONT SOLDERPASTE	String	SOLDERPASTE	100000410-001_Rev1_1_PASTE.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100000410-001_Rev1_1_CU.GBR	1.50	String
1700R_PP1520	String	PREPREG	String	0.10	
1700R_PP1116	String	PREPREG	String	4.00	4.14
INNER 1 COPPER	2	1 OZ COPPER	100000410-001_Rev1_1_CU.GBR	1.40	String
1700R_461028	String	PREPREG	String	10.00	4.00
INNER 2 COPPER	3	1 OZ COPPER	100000410-001_Rev1_1_CU.GBR	1.40	String
1700R_PP1116	String	PREPREG	String	4.70	4.23
1700R_PP1520	String	PREPREG	String	0.10	4.16
BACK COPPER	4	0.5 OZ COPPER	100000410-001_Rev1_1_CU.GBR	1.50	String
BACK SOLDERPASTE	String	SOLDERPASTE	100000410-001_Rev1_1_PASTE.GBR	1.00	4.50
BACK SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_1_SOLDERMASK.GBR	String	String
				TOTAL THICKNESS	42.30 +/-0.5
				BOARD OUTLINE	100000410-001_Rev1_1_OUTLINE.GBR
				PLATED THRU HOLES	100000410-001_Rev1_1PTH.DRL
				NON-PLATED THRU HOLES	100000410-001_Rev1_1NPTH.DRL

NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

NET TIE SHORTS				
NETTED PAIR	X PHS (MIL)	Y PHS (MIL)	LAYER	NETS SHORTED
HT1	7777.00	1100.00	01a	000A_000
HT2	1563.00	1200.00	12a	000B_000

IMPEDANCE					
TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
1	2	50	6.4	6.5	DIFF COPLANAR WAVEGUIDE W/COAD



Drill Map:

- 0.152mm / 0.0060" (172 holes)
- 0.813mm / 0.0320" (9 holes)
- + 1.448mm / 0.0570" (6 holes) (not plated)
- ◻ 2.200mm / 0.0866" (2 holes) (not plated)

NOTES, UNLESS OTHERWISE SPECIFIED:

- STANDARDS:
 - 1.1: THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-4012 CLASS 2.
 - 1.2: INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
 - 1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.3.22.
 - 1.4: DO NOT SCALE TO DRAWING.
- MATERIAL:
 - 2.1: THE PCB DIELECTRIC MATERIAL SHALL BE STIBO.
 - 2.2: THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
- FLATNESS:
 - 3.1: BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
 - 3.2: TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
- EDGE GEOMETRY:
 - 4.1: MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
 - 4.2: FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
- SURFACE FINISH:
 - 5.1: FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
- HOLES:
 - 6.1: SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
 - 6.2: PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-4012, CLASS 2.
 - 6.3: SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
 - 6.4: ALL HOLES SHALL BE LOCATED WITHIN 5 MIL OF THEIR POSITION AS SUPPLIED IN CAD DATA.
 - 6.5: THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
 - 6.6: HOLES IN THE HOLE TABLE MARKED AS (00q000)CAPPED/FILLED(00q000) SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
- SOLDERMASK:
 - 7.1: SOLDERMASK OVER BARE COPPER (P100C) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ARTWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
 - 7.2: COLOR: GREEN.
 - 7.3: LIQUID PHOTO-IMAGABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS; HALOGEN FREE.
 - 7.4: NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
- WELDING:
 - 8.1: SOLDERBOND PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-AUTITENT EPOXY INK.
 - 8.2: COLOR: WHITE.
 - 8.3: ANY UNSPECIFIED STRAKE WIDTH SHALL BE 5 MIL.
 - 8.4: CLIP SOLDERBOND AWAY FROM ANY EXPOSED METAL.
- FINISHING:
 - 9.1: REMOVE ALL BURRS AND BREAK SHARP EDGES.
- IMPEDANCE (ALL TOLERANCED +/- 10%)
 - 10.1: SEE IMPEDANCE REFERENCE TABLES.
 - 10.2: VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
 - 10.3: VENDOR MAY SUBSTITUTE SPECIFIC CORES OR PREPREGS TO ACHIEVE IMPEDANCE TARGETS AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
- PANELIZED PWB LAYOUT:
 - 11.1: IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBs TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
- NET TIES AND SHORTS:
 - 12.1: CONSULT THE NET TIE SHORTS TABLE FOR NETS THAT ARE INTENTIONALLY SHORTED WITH NET TIES.
- NON-DESTRUCTIVE EVALUATION:
 - 13.1: ALL PWBs SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-385 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
 - 13.2: CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
 - 13.3: ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS "ET". MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.

LAYER STACKUP AND FILENAME					
LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mil)	Er
FRONT SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_13_SOLDERMASK.GBR	String	String
FRONT SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_13_SOLDERMASK	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100000410-001_Rev1_13_C01R	1.50	String
1700R_PP2120	String	PREPREG	String	0.10	
1700R_PP2116	String	PREPREG	String	4.00	4.14
INNER 1 COPPER	2	1 OZ COPPER	100000410-001_Rev1_13_C02R	1.40	String
1700R_46102	String	PREPREG	String	10.00	4.00
INNER 2 COPPER	3	1 OZ COPPER	100000410-001_Rev1_13_C03R	1.40	String
1700R_PP2116	String	PREPREG	String	4.70	4.22
1700R_PP2120	String	PREPREG	String	0.10	4.26
BACK COPPER	4	0.5 OZ COPPER	100000410-001_Rev1_13_C04R	1.50	String
BACK SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_13_S04R.GBR	1.00	4.50
BACK SOLDERMASK	String	SOLDERMASK	100000410-001_Rev1_13_S04R01R.GBR	String	String
				TOTAL THICKNESS	42.20 +/-0.5
BOARD OUTLINE			100000410-001_Rev1_13_OU_13011.GBR		
PLATED THRU HOLES			100000410-001_Rev1_PTH1.DRL		
NON-PLATED THRU HOLES			100000410-001_Rev1-NPTH1.DRL		

NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

NET TIE SHORTS				
NETTED PAIR	X POS (MIL)	Y POS (MIL)	LAYER	NETS SHORTED
NET1	7777.00	1100.00	01a	000A_000
NET2	1563.00	1200.00	12a	000B_000

IMPEDANCE					
TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
1	2	50	6.4	6.5	DIFF COPLANAR WAVEGUIDE W/COAD